Final Capstone Project: Azure Databricks Healthcare Analytics Platform

Business Scenario

A hospital group wants to build an analytics platform using **Azure Databricks** to analyze patients, doctors, appointments, billing, departments, hospital preferences, revenue, and appointment trends.

Datasets Required

1. patients.csv

```
patient_id,patient_name,city,state,age,gender,insurance_status
P101,Rahul Sharma,Hyderabad,Telangana,35,Male,Active
P102,Priya Reddy,Bangalore,Karnataka,29,Female,Active
P103,Amit Kumar,Mumbai,Maharashtra,42,Male,Inactive
P104,Sneha Patel,Delhi,Delhi,31,Female,Active
P105,Farhan Ali,Chennai,Tamil Nadu,55,Male,Active
P106,Neha Singh,Pune,Maharashtra,38,Female,Inactive
P107,Arjun Verma,Hyderabad,Telangana,26,Male,Active
P108,Meera Nair,Kochi,Kerala,48,Female,Active
```

2. doctors.csv

```
doctor_id,doctor_name,department,city,consultation_fee
D101,Dr. Ramesh,Cardiology,Hyderabad,1500
D102,Dr. Priya,Neurology,Bangalore,2000
D103,Dr. Anita,Dermatology,Chennai,1000
D104,Dr. Suresh,Orthopedics,Mumbai,2500
D105,Dr. Meera,Pediatrics,Delhi,1200
D106,Dr. Kiran,Cardiology,Hyderabad,3000
```

3. appointments.csv

```
appointment_id,patient_id,doctor_id,appointment_date,diagnosis,bill_amour
A1001,P101,D101,2026-06-01,Heart Checkup,5000,Completed
A1002,P102,D102,2026-06-01,Migraine,3500,Completed
A1003,P103,D103,2026-06-02,Skin Allergy,2000,Pending
A1004,P104,D104,2026-06-02,Fracture,12000,Completed
A1005,P105,D105,2026-06-03,Fever,1500,Completed
A1006,P106,D106,2026-06-03,Heart Checkup,7000,Completed
A1007,P107,D101,2026-06-04,Chest Pain,5500,Completed
A1008,P108,D103,2026-06-04,Skin Infection,2500,Pending
A1009,P101,D106,2026-06-05,Cardiac Review,6500,Completed
A1010,P104,D104,2026-06-05,Back Pain,4500,Cancelled
```

4. patient_preferences.json

```
[
  {
    "patient_id": "P101",
    "preferred_hospital": "Apollo Hospital",
    "contact": {
      "phone": "9876500011",
      "email": "rahul@mail.com"
    }
  },
  {
    "patient_id": "P102",
    "preferred_hospital": "Yashoda Hospital",
    "contact": {
      "phone": "9876500012",
      "email": "priya@mail.com"
    }
  },
  {
    "patient_id": "P104",
    "preferred_hospital": "Care Hospital",
    "contact": {
      "phone": "9876500014",
      "email": "sneha@mail.com"
    }
```

```
  },
  {
    "patient_id": "P108",
    "preferred_hospital": "Apollo Hospital",
    "contact": {
      "phone": "9876500018",
      "email": "meera@mail.com"
    }
  }
]
```

Capstone Requirements

Part 1: Data Ingestion

```
1. Upload CSV and JSON files into Databricks.
2. Read patients.csv.
3. Read doctors.csv.
4. Read appointments.csv.
5. Read patient_preferences.json.
6. Display schema of all datasets.
7. Save raw data as Bronze Delta tables.
```

Part 2: Data Cleaning and Transformation

```
8. Handle missing values if present.
9. Flatten patient_preferences.json.
10. Join patients with preferences.
11. Join appointments with patients.
12. Join appointments with doctors.
13. Create final_bill = bill_amount + consultation_fee.
14. Create appointment_month from appointment_date.
15. Create patient_age_group:
    age >= 50 -> Senior
    age >= 30 -> Adult
    Else -> Young
16. Save transformed data as Silver Delta tables.
```

Part 3: Spark SQL

```
17. Create temporary views.
18. Write SQL query for total hospital revenue.
19. Write SQL query for revenue by department.
20. Write SQL query for revenue by city.
21. Write SQL query for completed appointments only.
22. Write SQL query for top patients by billing.
```

Part 4: Window Functions

```
23. Rank doctors by revenue.
24. Rank departments by revenue.
25. Find top 3 patients by billing.
26. Find top doctor in each department.
27. Create running revenue by appointment_date.
```

Part 5: Delta Lake

```
28. Create Delta table using DataFrame write.
29. Create Delta table using saveAsTable().
30. Create Delta table using SQL.
31. View Delta history.
32. Read previous Delta version using Time Travel.
33. Perform SCD Type 1 merge using updated patient city/insurance data.
34. Verify version increment after merge.
35. Run OPTIMIZE.
36. Run ZORDER BY(patient_id).
37. Run VACUUM.
```

Part 6: Visualization

```
38. Bar chart: Revenue by Department.
39. Bar chart: Revenue by City.
40. Pie chart: Appointment Status Distribution.
```

41. Horizontal bar chart: Top Doctors by Revenue.
42. Line chart: Revenue Trend by Date.

Part 7: Tables and Views

43. Create managed table.
44. Create external table.
45. Create temporary view.
46. Create global temporary view.
47. Compare scope and lifetime of each.